higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 4

(6050394)

13 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Horizontal pulpers ...

A do not have a rotor.
B are applied as under-machine broke pulpers.
C are used to slush stock that requires deinking.
D have a better efficiency than vertical pulpers.

1.2 It can be said that the thickening process is functioning poorly if …

A the water temperature cannot be controlled.
B water has a high dissolved solids content.
C the water has a milky colour.
D suspended solids are still present in the water.

1.3 … prevents stock from spilling over the edge of the Fourdrinier wire.

A Couch roll
B Dandy roll
C Table roll
D Rubber deckle

1.4 … refers to how much light is reflected by the surface of a paper sheet.

A Brightness
B Lightness
C Opacity
D Whiteness

1.5 Which factor affects drainage at the forming board?

A Vacuum level
B Slice geometry
C Suction couch position
D Forming board size

1.6 The most important process quality requirement for made paper is …

A smoothness.
B formation.
C freeness.
D porosity.

1.7 During papermaking, a lot of energy is consumed at the ...

A press section.
B forming section.
C drying section.
D winder section.

1.8 Yankee dryers can be …

A as large as 5,5 m in diameter and weigh up to 180 tons.
B as large as 4,5 m in diameter and weigh up to 100 tons.
C as large as 4 m in diameter and 3 m wide.
D as large as 5,5 m in diameter and 5 m wide.

1.9 Black liquor oxidation …

A increases the generation of the hydrogen sulphide.
B reduces corrosion in the lime kiln.
C increases lime consumption.
D reduces sulphur losses in the direct contact evaporator.

1.10 The calcining reaction is represented as:

A $CaCO_3$ + heat → $CaO + CO_2$
B Ca_2CO_3 + heat → $Ca_2O + CO_2$
C $CaO + H_2O \rightarrow Ca(OH)_2$ + heat
D $Ca(OH) + Na_2CO_3 \rightarrow 2NaOH + CaCO_3$

(10 × 1) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK. Correct the statement if it is FALSE.

2.1 The lower the Canadian Standard Freeness (CSF) test value, the freer the pulp.

2.2 Sheet formation starts with a low drainage in order to prevent fibre loss.

2.3 Rewinding is used to remove certain quality paper defects.

2.4 Sulphite pulping makes use of acidic sulphite cooking liquor.

2.5 Paper quality with wet and dry streaks indicates poor formation.

(5 × 1) **[5]**

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QUESTION 3

Choose a description from COLUMN B that matches a word or a term in COLUMN A. Write only the letter (A–L) next to the question number (3.1–3.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	Sulphur dioxide gas	A	can be recovered for re-use
3.2	Refining	B	mixing of green liquor with lime to form a slurry
3.3	Fluid shear	C	material that can withstand extremely high temperatures
3.4	Yankee	D	is bled off to regulate pressure in the digester
3.5	Web control	E	used to dry low-weight paper products
3.6	Black liquor	F	mechanically conditions the pulp fibres to improve pulp properties
3.7	Refractory	G	combusted in a recovery furnace
3.8	Heavy black liquor	H	used to impart important property of wound paper
3.9	NSSC liquor	I	force experienced by an object when a fluid flows over it
3.10	Slaking	J	more corrosive than kraft liquor
		K	used as conduit for air drying
		L	needed for oxidation of black liquor

(10 × 1) **[10]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

4.1 Give the FIVE steps involved in waste paper processing. (5)

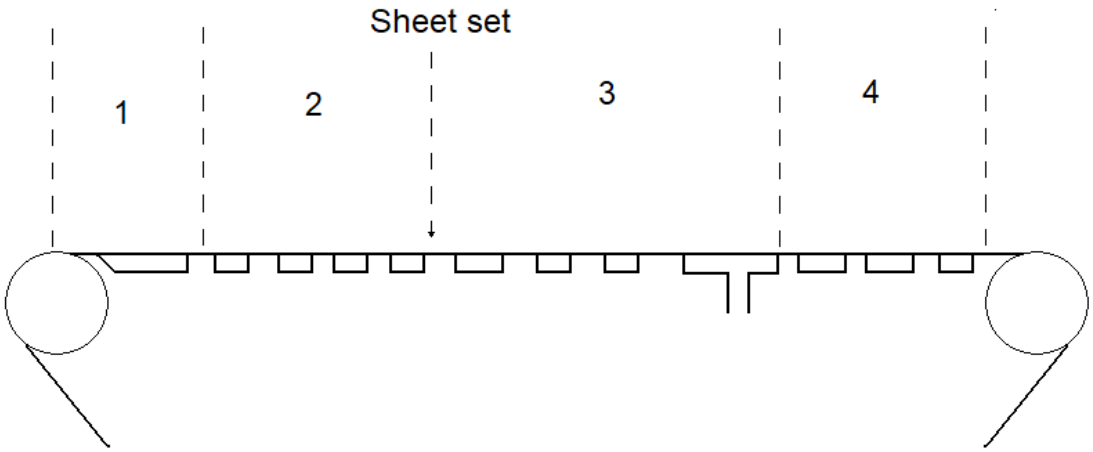
4.2 How would you determine the papermaking potential of pulp? Elaborate on THREE key steps. (6)

4.3 What flake content should pulp generally have? (1)

[12]

QUESTION 5

5.1 Study the following Fourdrinier table diagram and answer the questions.



5.1.1 Name the FOUR phases on the forming table above. (4)

5.1.2 Describe what happens in each phase mentioned in QUESTION 5.1.1. (4 × 2) (8)

[12]

QUESTION 6

6.1 6.1.1 Draw and label a neat sketch of a rate of evaporation graph. (4)

6.1.2 Name THREE operational problems that will occur with the quality of paper when the temperature of the cylinders is too high during the heating stages of drying. (3)

6.2 List the FOUR factors that are related to web control during winding. (4)

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6.3 State ONE corrective action for each of the following causes of winder wrinkles defect:

6.3.1 Low sheet tension

6.3.2 Sticking core shafts

6.3.3 Misaligned rolls

6.3.4 Rolls stretching and overlap

(4 × 1) (4)
[15]

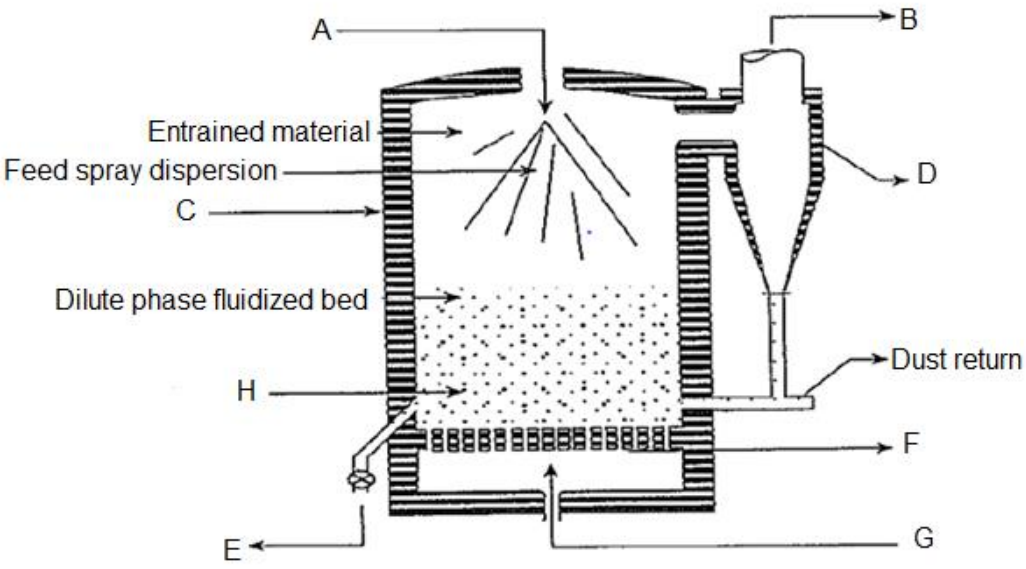
QUESTION 7

7.1 What is black liquor? (1)

7.2 State THREE advantages of burning off the organic material in black liquor. (6)

7.3 Explain how the solids concentration of black liquor affects the quality of heavy black liquor fired into a furnace. (6)

7.4 Study the diagram below and answer the questions.



7.4.1 Provide labels for the above diagram (A to H). (8)

7.4.2 Compare the state of the final product produced by the Copeland reactor above to the one produced by the Kraft furnace. (2)

7.4.3 What is used to fluidise the bed and what is the disadvantage of using a fluidised bed reactor? (2)
[25]

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QUESTION 8

8.1 Answer the following questions with regards to chemical production using a lime kiln:

8.1.1 State ONE problem that would occur when pellets aggregate to large balls in the lime kiln. (2)

8.1.2 Give the THREE phases that are involved in a typical calcinations process. (3)

8.2 Provide TWO reasons why calcium bisulphite is not recovered. (2)

8.3 State FOUR tests that can be performed on unslaked lime. (4)

[11]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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